Project Report: Library Management System (LMS)

Project Overview

The Library Management System (LMS) is a web-based application designed to manage library resources, including books, sections, and user interactions. The system is built using Flask for the backend, SQLite for data storage, and integrates JWT-based authentication for secure access control. The application allows librarians to manage books, sections, and book requests, while users can request books, view their requests, and provide feedback.

Technologies Used

- **Flask**: Micro web framework used to build the backend.
- **SQLAlchemy**: ORM for database interactions.
- **JWT (JSON Web Tokens)**: For authentication and authorization.
- **SQLite**: Lightweight database for storing application data.
- **Werkzeug**: For handling file uploads and secure file naming.
- **bcrypt**: For password hashing.
- **Flask-CORS**: To handle Cross-Origin Resource Sharing.
- **Flask-JWT-Extended**: For JWT-based authentication.

Key Features

1. User Management

- **Registration**: Users can register with a username, password, and role (librarian or user). Passwords are securely hashed using bcrypt.
- **Login**: Users can log in using their credentials to receive a JWT for authenticated access.
- **Permission Management**: Users have a permission_granted_at attribute to track permission validity for requesting books.

2. Librarian Features

- **Dashboard**: Provides an overview of sections and books in the library.
- **Manage Sections**: Librarians can add, update, and delete library sections.
- **Manage Books**: Librarians can add, update, and delete books. PDFs for books are uploaded and stored on the server.

- **Manage Book Requests**: Librarians can view, accept, or deny book requests. Requests can be tracked with their status and days since acceptance.

3. User Features

- **User Dashboard**: Displays user's book requests, available sections, and books. Users can request books from available sections.
- **Book Requests**: Users can request books and view the status of their requests.
- **Feedback**: Users can provide feedback on books they have read, including rating and comments.

4. File Handling

- **PDF Uploads**: Books can be uploaded with associated PDF files, which are stored in the uploads directory on the server.

API Endpoints

User Endpoints

- **/register**: Registers a new user.
- **/login**: Authenticates a user and returns a JWT.
- **/user/dashboard**: Provides user-specific dashboard data.
- **/user/request_book/<int:book_id>**: Allows users to request a book.

Librarian Endpoints

- **/librarian/dashboard**: Displays sections and books.
- **/librarian/add_section**: Adds a new section.
- **/librarian/add_book**: Adds a new book, including PDF upload.
- **/librarian/delete_book/<int:book_id>**: Deletes a book and its associated PDF file.
- **/librarian/update_book/<int:book_id>**: Updates book details.
- **/librarian/delete_section/<int:section_id>**: Deletes a section.
- **/librarian/sections**: Lists all sections.
- **/librarian/section/<int:section_id>**: Retrieves a specific section.
- **/librarian/update_section/<int:section_id>**: Updates section details.
- **/librarian/book_requests**: Lists all book requests.
- **/librarian/accept_request/<int:request_id>**: Accepts a book request.
- **/librarian/deny_request/<int:request_id>**: Denies a book request.

File Handling

- **/uploads/<filename>**: Serves the PDF files of books.

Database Schema

User Table

- id: Integer, primary key.
- username: String, unique.
- password: String.
- role: String.
- permission_granted_at: DateTime.

Section Table

- id: Integer, primary key.
- name: String.
- description: String.

Book Table

- id: Integer, primary key.
- name: String.
- content: Text.
- authors: String.
- section_id: Integer, foreign key to section.
- date_issued: DateTime.
- return_date: DateTime.
- pdf_file: String.
- content_text: Text.

BookRequest Table

- id: Integer, primary key.
- user_id: Integer, foreign key to user.
- book_id: Integer, foreign key to book.
- timestamp: DateTime.
- status: String.

Feedback Table

- id: Integer, primary key.
- user_id: Integer, foreign key to user.
- book_id: Integer, foreign key to book.
- rating: Integer.
- comment: Text.
- timestamp: DateTime.

Security Considerations

- **JWT Authentication**: Secures access to endpoints by requiring a valid token.
- **Password Hashing**: Passwords are hashed using bcrypt to ensure security.
- **Role-Based Access Control**: Different endpoints are protected based on user roles (librarian vs. user).

Link to the presentation video: nxm-wsto-hzq (2024-08-12 06:34 GMT+5:30)